

Topic Modeling for Text Data

From LDA to Structural Topic Models

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Today's Question

Suppose we have thousands of documents.

How can we discover what people are talking about without reading every document?

Examples:

- open-ended survey responses
- political speeches
- research abstracts
- news articles
- social media posts

Learning Goals

By the end of the session, you should be able to:

- 1 Explain what a topic model is trying to estimate.
- 2 Construct a document-term matrix.
- 3 Interpret the LDA plate diagram.
- 4 Fit and interpret a basic LDA model in R.
- 5 Explain why Structural Topic Models extend LDA.
- 6 Use document metadata to ask regression-style questions about topics.

Two-Hour Plan

Time	Segment	Mode
0–15	Text as data + warm-up exercise	discussion
15–30	Latent variables and topics	presentation
30–50	LDA plate diagram and generative story	presentation
50–65	Preprocessing and DTM	R exercise
65–80	Fit LDA in R	demonstration
80–95	Interpreting topics	discussion
95–105	Why LDA is incomplete	presentation
105–120	Structural Topic Models	demo + exercise

Basic idea

A topic model treats each document as a mixture of latent themes called **topics**.

Document

The economy added jobs, but inflation remains a concern.

Possible topic mixture

- Economy: 70%
- Labor: 20%
- Politics: 10%

Warm-Up Exercise: Discovering Topics by Hand

Instructions: Working in pairs, read the eight mini-abstracts and identify the major themes.

- 1 We investigate voting patterns in congressional elections using demographic and economic indicators.
- 2 We develop a Bayesian model for predicting hospital readmission rates.
- 3 We study climate change impacts on agricultural productivity across the Midwest.
- 4 We propose a deep learning approach for medical image segmentation.
- 5 We examine the relationship between air pollution and respiratory health.
- 6 We analyze public attitudes toward environmental policy using survey data.
- 7 We develop statistical methods for high-dimensional genomic data.
- 8 We investigate machine learning techniques for climate forecasting.

Discussion: What Just Happened?

Before we talk about topic models, consider the questions you just answered:

- ① How many topics are present?
- ② Does every document belong to exactly one topic?
- ③ Which documents seem to belong to multiple topics?
- ④ Could reasonable people disagree about the topics?

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Key idea

Humans infer hidden topics from words. Topic models treat those topics as *latent variables* and estimate them probabilistically.

What Is a Latent Variable?

A latent variable is a quantity that is not directly observed but is inferred from observed data.

Examples:

- intelligence inferred from test scores
- disease status inferred from symptoms
- customer satisfaction inferred from survey responses
- topic inferred from word usage

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In topic modeling:

Words observed $\implies$ Topics latent

LDA says:

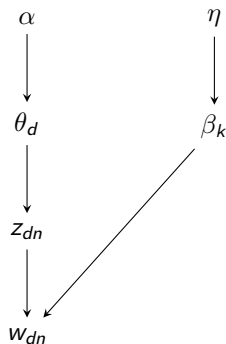
Documents are mixtures of topics.
Topics are mixtures of words.

The model tries to infer both mixtures from the observed words.

LDA Notation

Symbol	Meaning	Observed?
w_{dn}	word n in document d	yes
z_{dn}	topic assignment for word n in document d	no
θ_d	topic mixture for document d	no
β_k	word distribution for topic k	no
α	prior parameter for document-topic mixtures	fixed / estimated
η	prior parameter for topic-word distributions	fixed / estimated

LDA Plate Diagram



Observed

- w_{dn} : observed word

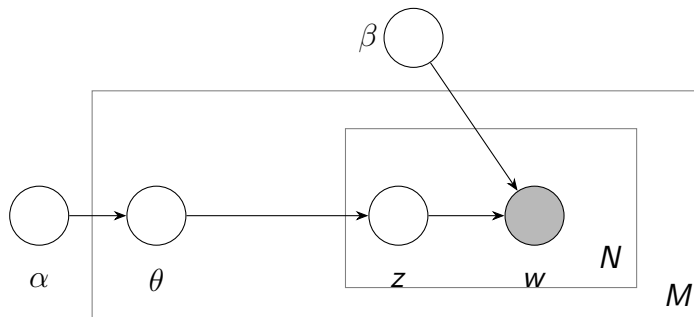
Latent

- θ_d : topic mixture
- z_{dn} : topic assignment
- β_k : topic-word distribution

Read arrows as:

- $\theta_d \rightarrow z_{dn}$
- $\beta_k \rightarrow w_{dn}$
- $z_{dn} \rightarrow w_{dn}$

LDA Plate Diagram: Formal Version



Source: Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). *Latent Dirichlet Allocation*. *Journal of Machine Learning Research*, 3, 993–1022.

What Are We Trying to Estimate?

We observe only the words in each document.

Quantity	Interpretation	Why we care
θ_d	topic mixture for document d	summarizes documents
β_k	word distribution for topic k	helps label topics
z_{dn}	topic assignment for word n	links words to topics

Inference problem

Given only the observed words w_{dn} , can we estimate the hidden quantities?

The Generative Story

For each topic:

$$\beta_k \sim \text{Dirichlet}(\eta)$$

For each document:

$$\theta_d \sim \text{Dirichlet}(\alpha)$$

For each word position:

$$z_{dn} \sim \text{Categorical}(\theta_d)$$

$$w_{dn} \sim \text{Categorical}(\beta_{z_{dn}})$$

Connecting the Diagram to the Probability Model

$$P(w, z, \theta, \beta) = \prod_k P(\beta_k) \prod_d P(\theta_d) \prod_n P(z_{dn} \mid \theta_d) P(w_{dn} \mid z_{dn}, \beta)$$

Takeaway

Each arrow in the plate diagram corresponds to a conditional distribution in the joint model.

Two Kinds of Probabilities

Document-topic probabilities

Topic	Probability
Economy	0.60
Health	0.25
Foreign policy	0.15

Topic-word probabilities

Word	Probability
jobs	0.08
inflation	0.07
wages	0.05
market	0.04

From Text to Data

To fit the model, we need to transform raw text into a form the computer can use.

Common steps:

- tokenize documents into words
- remove stop words
- remove punctuation, numbers, or rare words
- optionally stem or lemmatize words
- create a document-term matrix

Important warning

Preprocessing choices are modeling choices. They can change the topics you discover.

Document-Term Matrix

Document	economy	jobs	health	vote	climate
Doc 1	2	1	0	0	0
Doc 2	0	0	3	0	0
Doc 3	1	0	0	2	1
Doc 4	0	0	0	1	4

Rows are documents. Columns are words. Entries are word counts.

R Exercise: Tokenize Text

```
library(tidyverse)
library(tidytext)

documents <- read_csv("abstracts.csv")

tokens <- documents %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word")

tokens %>%
  count(word, sort = TRUE)
```

Exercise:

- What are the most common words?
- Are there words you would add to a custom stop-word list?

Preprocessing: From Text to a Document-Term Matrix

```
documents <- read_csv("abstracts.csv")

tokens <- documents %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word")

word_counts <- tokens %>%
  count(id, word)

dtm <- word_counts %>%
  cast_dtm(document = id,
           term = word,
           value = n)
```

Raw Text → Tokenization → Remove Stop Words → Word Counts → DTM

Fitting an LDA Topic Model

Goal: Discover latent topics from the document-term matrix.

```
library(topicmodels)

# Fit a topic model with 5 topics
lda_fit <- LDA(
  dtm,
  k = 5,
  control = list(seed = 1234)
)

# Display the 10 most probable
# words for each topic
terms(lda_fit, 10)
```

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Discussion

- Why $k = 5$?
- What if $k = 3$?
- What if $k = 10$?

Visualize Top Words by Topic

```
beta <- tidy(lda_fit, matrix = "beta")

beta %>%
  group_by(topic) %>%
  slice_max(beta, n = 10) %>%
  ungroup() %>%
  ggplot(aes(beta, reorder_within(term, beta, topic))) +
  geom_col() +
  facet_wrap(~ topic, scales = "free") +
  scale_y_reordered()
```

What LDA Ignores

Suppose we know something about each document.

Examples:

- year written
- political party
- country
- author gender
- treatment group
- publication source

Where does that information enter the LDA model?

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Nowhere.
LDA only sees words.

Where does that information enter the LDA model?

The Statistical Problem

LDA can tell us what topics appear in documents.

But researchers often want to ask:

- Are some topics more common among one group than another?
- Do topics become more or less prevalent over time?
- Does an experimental treatment affect what people write about?
- Do different groups use different words when discussing the same topic?

These are regression-style questions.

STM extends LDA

Structural Topic Models allow document-level metadata to influence topic prevalence and, optionally, topic content.

Two major uses of metadata:

- **Topic prevalence:** which topics are more common?
- **Topic content:** what words are used within a topic?

STM as a Regression Extension

In LDA:

$$\theta_d \sim \text{Dirichlet}(\alpha)$$

In STM, conceptually:

$$\theta_d = f(X_d)$$

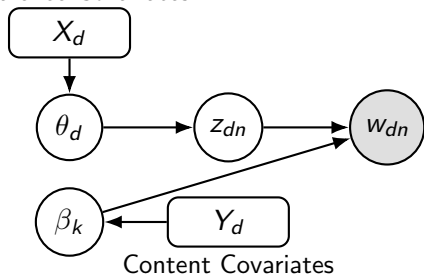
where X_d contains document-level covariates.

Examples:

$$X_d = \{\text{party, year, treatment group}\}$$

Conceptual STM Diagram

Prevalence Covariates



Interpretation

- X_d : affects which topics appear in a document.
- Y_d : affects how topics are expressed through words.

STM: Incorporating Document Metadata

```
library(readr)
library(stm)

abstracts <- read_csv("abstracts.csv")

# Add document metadata (I made this up)
abstracts$field <- c(
  "policy", "health",
  "environment", "health",
  "environment", "policy",
  "statistics",
  "environment"
)
```

```
processed <- textProcessor(
  documents = abstracts$text,
  metadata = abstracts
)
```

STM: Fitting the Model

```
out <- prepDocuments(  
  processed$documents,  
  processed$vocab,  
  processed$meta  
)  
  
stm_fit <- stm(  
  documents = out$documents,  
  vocab = out$vocab,  
  K = 5,  
  prevalence = ~ field,  
  data = out$meta,  
  seed = 1234  
)
```

Key Difference from LDA

LDA only uses:

- words

STM uses:

- words
- document metadata

STM: Fitting the Model

```
out <- prepDocuments(  
  processed$documents,  
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```

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LDA only uses:

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STM uses:

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Important Line

```
prevalence = ~ field
```

This allows topic prevalence to depend on the document's field.

Estimate Effects of Metadata

```
effects <- estimateEffect(  
  1:5 ~ field,  
  stm_fit,  
  meta = out$meta  
)  
  
summary(effects)  
plot(effects, covariate = "field", topics = c(1, 3, 5))
```

Exercise:

- Choose one topic.
- Estimate whether its prevalence differs by metadata group.
- Interpret the result in ordinary language.

LDA vs. STM

Feature	LDA	STM
Uses words	yes	yes
Latent topics	yes	yes
Document-topic mixtures	yes	yes
Metadata affects topic prevalence	no	yes
Metadata affects topic content	no	optional
Regression-style interpretation	limited	central

Choosing the Number of Topics (K)

LDA

Fit models for several values of K .

```
result <- FindTopicsNumber(  
  dtm,  
  topics = 2:10  
)  
FindTopicsNumber  
plot( result)
```

Evaluate:

- Semantic coherence
- Topic interpretability

STM

Use built-in diagnostics.

```
k_result <- searchK(  
  documents =  
    out$documents,  
  vocab =  
    out$vocab,  
  K = c(3,5,7,10)  
)  
plot(k_result)
```

Evaluate:

- Semantic coherence
- Exclusivity

Choosing the Number of Topics

Statistical criteria can help narrow the range of reasonable values, but interpretability and scientific usefulness still matter.

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Diagnostic	Question
Semantic Coherence	Do the top words tend to occur together in documents?
Exclusivity	Do the top words distinguish this topic from the others?

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Diagnostic	Question
Semantic Coherence	Do the top words tend to occur together in documents?
Exclusivity	Do the top words distinguish this topic from the others?

Important

There is rarely a single correct value of K . A useful topic model balances semantic coherence and exclusivity.

Interpretation Checklist

For any topic model, ask:

- 1 Are the top words coherent?
- 2 Can we give the topic a meaningful name?
- 3 Which documents are most associated with the topic?
- 4 Is the topic stable across reasonable choices of K ?
- 5 Are metadata effects substantively meaningful?
- 6 What preprocessing choices might have affected the result?

Mini-Project

In small groups using either the folktale data or the nsf_abstracts data:

- 1 Fit an LDA or STM model.
- 2 Choose a value of K and justify it.
- 3 Name 3–5 topics.
- 4 Identify one interesting document for a topic.
- 5 If using STM, interpret one metadata effect.

Prepare a two-minute report:

What did the model reveal, and what should we be cautious about?

Closing Thought

Topic models are not machines that discover truth.

They are statistical tools for organizing, summarizing, and questioning large collections of text.

Research Requires Judgment

The analyst still has to choose the model, evaluate the results, and justify the conclusions.